

EEG WORKSHOP: Investigating electroencephalographic signals through different analytic paradigms

Damián Dellavale^{1,2}, Sebastián Moguilner^{2,3}, Carlos Coronel^{2,4}, Jhosmary Cuadros⁴,
Pavel Prado⁵, Agustín Ibañez^{2,4}

¹ Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET), Centro Atómico Bariloche, Argentina.

² Global Brain Health Institute (GBHI), Trinity College Dublin (TCD), Dublin, Ireland.

³ Medical Biodynamic Center, Massachusetts General Hospital, Harvard Medical School, USA.

⁴ Latin American Brain Health Institute (BrainLat), Universidad Adolfo Ibañez, Santiago de Chile, Chile.

⁵ Universidad San Sebastián, Santiago de Chile, Chile.

Contact: damian.dellavale@gbhi.org (Damián Dellavale)

Objectives

A three-module intensive workshop tailored to the datasets, experimental protocols, and research priorities of the EEG laboratories at *Istanbul Medipol University*, providing hands-on training in the analysis of oscillatory and arrhythmic activities, avalanche-like events and EEG-informed biological brain age. Each module integrates a dedicated segment on emerging research frontiers, fostering the identification of concrete opportunities for scientific collaboration.

Program

- *Oscillations in EEG signals.*
Filtering. Hilbert transform, Circular statistics. **Analytic examples.**
Coherence measures (Spectral coherence, PLV, wPLV, PPC, PLI, wPLI).
Cross frequency couplings (PAC, PFC, PPC). Cross-Frequency Directionality (CFD). Mechanistic interpretations (Theta-Gamma code, Gamma-leading Theta interactions). **Coding examples.**
Phase Clustering (PC). **Coding examples.**
Spectral harmonicity (TLI). **Coding examples.**
Statistical analysis: Nonparametric permutation testing. **Coding examples.**
Ideas for collaboration: CFC directionality. CFC and functional networks. CFC and working memory. CFC and neuromodulation. Assessing cortical excitability using EEG and tES. Transcranial Temporal Interference Stimulation (tTIS).
- *Broadband arrhythmic activity in EEG signals.*
Neuronal avalanches and criticality hypothesis. Aperiodic and scale-free activities. Long-range temporal correlations. Detrended fluctuation analysis scaling exponent and spectral aperiodic exponent.
Linking oscillations and transient salient events: Spectral group delay consistency. **Analytic and Coding examples.**
Avalanche-like events in EEG as a biomarker of neurological disorders. **Coding examples for Parkinson's disease.**
Ultradian dynamics of salient events.
Ideas for collaboration: Ultradian dynamics of salient events in brain disorders. Avalanche-like events and neuromodulation.
- *Brain Aging Models.*
Normative modeling based on neuroimaging.
Brain aging charts: Brain feature ~ Chronological age + covariates
Brain Clock Models: Biological brain age ~ Brain feature(s)
Biological age vs. chronological age.
Functional vs. structural connectivity. Power, phase, group delay and O-info based connectivity.
Regression model for building the clock. Methodological caveats.
Non-invasive biophysical neuromodulation to reduce functional brain aging. **Coding examples.**
Ideas for collaboration: Brain clocks based on arrhythmic activity. Brain clocks and neuromodulation.

Resources

https://github.com/damian-dellavale/EEG_workshop

Schedule

3 August 2026: Oscillations in EEG signals

10:00 - 11:30: Filtering. Hilbert transform. Coherence measures.

11:30 - 11:45: Coffee Break.

11:45 - 13:15: Coherence measures.

13:15 - 14:15: Lunch Break.

14:15 - 15:45: Cross-Frequency Coupling (CFC). Cross-Frequency Directionality (CFD). Mechanistic interpretations. Phase Clustering (PC). Spectral harmonicity (TLI).

15:45 - 16:00: Break.

16:00 - 17:00: Statistical analysis: Nonparametric permutation testing.

4 August 2026: Broadband arrhythmic activity in EEG signals

10:00 - 11:30: Neuronal avalanches and criticality hypothesis. Aperiodic and scale-free activities. Long-range temporal correlations. Detrended fluctuation analysis scaling exponent and spectral aperiodic exponent.

11:30 - 11:45: Coffee Break.

11:45 - 13:15: Linking oscillations and transient salient events: Spectral group delay consistency.

13:15 - 14:15: Lunch Break.

14:15 - 15:45: Avalanche-like events in EEG as a biomarker of neurological disorders.

15:45 - 16:00: Break.

16:00 - 17:00: Ultradian dynamics of salient events.

5 August 2026: Brain Aging Models

10:00 - 11:30: Normative modeling based on neuroimaging. Brain aging charts.

11:30 - 11:45: Coffee Break.

11:45 - 13:15: Brain Clock Models. Biological age vs. chronological age.

13:15 - 14:15: Lunch Break.

14:15 - 15:45: Connectivity measures. Regression model for building the clock. Methodological caveats.

15:45 - 16:00: Break.

16:00 - 17:00: Non-invasive biophysical neuromodulation to reduce functional brain aging.

6 and 7 August 2026:

10:00 - 13:15: Q&A. Revisiting specific topics of interest. Hands-on coding. Collaboration ideas.

14:15 - 17:00: Q&A. Revisiting specific topics of interest. Hands-on coding. Collaboration ideas.